

Duc Hoang

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SUMMARY

PhD candidate in High-Energy Physics at MIT and CERN developing real-time machine learning systems for nanosecond-scale decision making. My work spans FPGA-optimized neural architectures, adaptive hardware-native learning algorithms, and large-scale statistical analyses for precision Higgs measurements. Recently, I've been extending these ideas to quantum-classical integration for scalable quantum technologies.

EDUCATION

- **Massachusetts Institute of Technology (MIT)** 2021 - Present
PhD, Physics. Thesis: Harder, Better, Faster, Stronger: Pushing the Data Frontier at the Large Hadron Collider Cambridge, USA
- **Rhodes College** 2017-2021
B.S., Physics, Summa Cum Laude. Thesis: Triggering on electron-nucleon scattering events at LDMX Memphis, USA

PROFESSIONAL EXPERIENCE

- **MIT & European Organization for Nuclear Research (CERN)** 2021 - Present
Graduate Researcher & Next Generation Trigger PhD Fellow Cambridge, USA & Geneva, Switzerland
 - Coordinated multi-institutional team to develop **ultra-low-latency FPGA-accelerated ML models** for CMS Level-1 Trigger, which processes approximately 10% of global internet traffic in real time. [Hear me explain it along the trigger leadership team](#) [4]
 - Built **first FPGA-based ML algorithms for real-time jet tagging** and delivered cross-platform, low-latency data transfer tests between different FPGA board platforms. [5]
 - Developed the **first sub-microsecond on-chip online learning system**, enabling model-free training at the timescale of physical processes like quantum control and fusion [2,3].
 - Analyzed all Run 2 CMS data (a few PBs) to deliver **most sensitive Higgs $\rightarrow b\bar{b}$ measurement in the $V(qq)H(bb)$ decay channel**, invited presentation at Rencontres de Moriond 2025. [6]
 - Mentored MIT undergraduate and master's theses related to CMS Level-1 Trigger, including **novel ML architectures for ultra-fast, efficient FPGA inference** [1,11].
 - Designed and led hands-on CERN trainings/hardware schools on ML for FPGA-based Level-1 Trigger applications; developed MIT 8.16/8.316 data-analysis projects published on MITx ([Computational Data Science in Physics](#)).
- **Espresso (MIT Sandbox Team)** 2025 - Present
Co-founder Cambridge, USA
 - Developing **efficient real-time and local data processing capabilities** for space and defense industry.
 - Defined the company's technical vision and led early customer discoveries focusing on space applications.
- **Fermi National Accelerator Laboratory** 2019 - 2021
Undergraduate Researcher Chicago, USA
 - Developer of [hls4ml](#), A package for **ultra-low-latency machine learning inference in FPGAs** using high level synthesis language (HLS). [5]
 - Designed superconducting magnet quench detection systems [6]
 - Analyzed simulation data to predict potential signal gains for the real-time trigger systems for Light Dark Matter Experiment (LDMX). [7]
 - Machine Learning analysis to infer hyper-parameter optimization performance of models used in the MINERvA Experiment [8].

SELECTED PUBLICATIONS & SOFTWARE

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| [1] D. Hoang et al. "KANELÉ: Kolmogorov-Arnold Networks for Efficient LUT-based Evaluation," <i>Proceedings of the 2026 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (ISFPGA'26)</i> , Seaside, CA, USA, 2026. Best Paper Nominee (Top 3 Paper)
https://arxiv.org/abs/2512.12850 | Kolmogorov-Arnold Networks," 2026. <i>Under review at ICML'26</i> .
https://arxiv.org/abs/2602.02056 |
| [2] D. Hoang . "Position: The Need for Ultrafast Training," <i>2nd Workshop on Domain-Specialized FPGAs (WDSFPGA'26)</i> , Seaside, CA, USA, 2026.
https://arxiv.org/abs/2602.02005 | [4] CMS Collaboration , "The Phase-2 Upgrade of the CMS Level-1 Trigger",
https://cds.cern.ch/record/2714892 |
| [3] D. Hoang et al. "Ultrafast On-chip Online Learning via Spline Locality in | [5] CMS Collaboration , 2025. "Multiclass object tagging in the CMS Phase-2 Level-1 Correlator Trigger." CMS Detector Performance Summary.
https://cds.cern.ch/record/2936315 . |

- [6] **CMS Collaboration**, "Search for a boosted Higgs boson decaying to bottom quark pairs in association with a hadronically decaying W or Z boson with the CMS detector using proton-proton collisions at $\sqrt{s} = 13$ TeV", <https://cds.cern.ch/record/2928061>
- [7] **hls4ml Community**, "hls4ml: An Open-Source Codesign Workflow to Empower Scientific Low-Power Machine Learning Devices"
- [8] **D. Hoang et al.**, "IntelliQuench: an adaptive machine learning system for detection of superconducting magnet quenches.," in *IEEE Transactions on Applied Superconductivity*, doi: 10.1109/TASC.2021.3058229.
- [9] **LDMX Collaboration**, "Current Status and Future Prospects for the Light Dark Matter eXperiment" [[arXiv:2203.08192](https://arxiv.org/abs/2203.08192)][[hep-ex](https://arxiv.org/abs/2203.08192)].
- [10] **D. Hoang et al.** "Inferring Convolutional Neural Networks' Accuracies from Their Architectural Characterizations," *2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA)*, Boca Raton, FL, USA, 2019, pp. 1388-1391. DOI: 10.1109/ICMLA.2019.00226
- [11] **CMS Collaboration**. 2024. "NNPuppiTaus: PUPPI Tau Reconstruction in the Level-1 Trigger with Real-Time Machine Learning." CMS Detector Performance Summary. <https://cds.cern.ch/record/2895660>.

MENTORSHIP & OUTREACH

Aarush Gupta, MIT BS/MS Thesis (CS & Physics). *Subsequent position: ML Engineer, Jane Street*
Orion Foo, MIT BS (CS & Physics).

- **SEAS - Summer in Engineering and Applied Sciences** 2025-present
Founder & General Manager 
 ◦ Founded, along with other researchers from Central Vietnam, a STEM [summer school](#), providing under-privileged but highly talented high school students in Central Vietnam the opportunity to study emerging fields such as artificial intelligence, data science, programming, renewable energy, and quantum technology at no cost.

HONORS AND AWARDS

- **The 2025 Breakthrough Prize in Fundamental Physics** 2025
Received as a member of the CMS Collaboration 
 ◦ The prize is awarded to physicists from theoretical, mathematical, or experimental physics that have made transformative contributions to fundamental physics, and specifically for recent advances.
- **2025 European Physical Society Conference on High Energy Physics (EPS-HEP) Poster Prize** 2025
European Physical Society (EPS) 
 ◦ Awarded for best poster at EPS-HEP 2025 (out of more than 100 posters), a highly prestigious biannual international conference featuring major results from leading high-energy physics experiments and collaborations.
- **MIT Public Service Fellow (PKG) Fellowship** 2025
Priscilla King Gray Public Service Center - MIT 
 ◦ Awarded to develop and implement a social impact project in collaboration with community partners in Central Vietnam.
- **Next Generation Trigger PhD fellow.** 2024
CERN 
 ◦ Awarded for exceptional PhD students to work along CERN scientists to develop the next generation trigger system at the Large Hadron Collider.
- **2024-25 Christiaan Huygens Graduate Fellowship - MIT.** 2024
MIT School of Science
 ◦ Fellowship providing financial support for the full academic year in the School of Science for students with non-traditional backgrounds.
- **MIT-The Royal Game of Ur Champion** 2023
MIT Playur II Tournament 
 ◦ The Royal Game of Ur is the oldest board game in the world.
- **Henry W Kendall (1955) first year graduate fellowship.** 2021
MIT Laboratory for Nuclear Science
 ◦ Financial support for first year graduate students.
- **Undergraduate Awards.** 2017-2021
Rhodes College
 ◦ Research Award in Physics, Council on Undergraduate Research (CUR) Physics & Astronomy Student Travel Award, W.O Shewmaker Award for Excellence in First-Year Search Program, Award for Excellence in Physics by a First-Year Student, Summa Cum Laude, Sigma Pi Sigma, Phi Beta Kappa, Physics Honors.